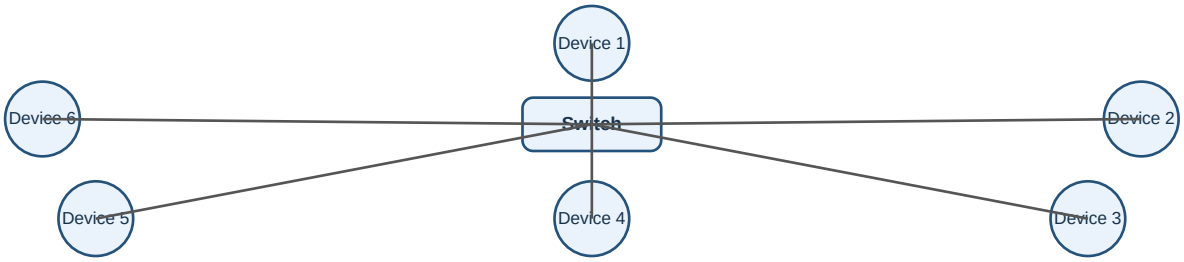
Network Topology Assignment

Name: _____ Class: _____

1. Star, Bus, and Mesh Diagrams with Examples

Star: All devices connect to a central switch.

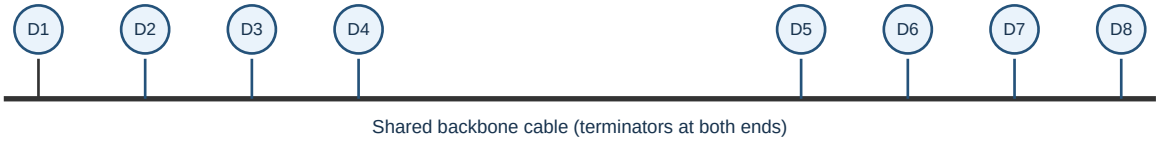
Star Topology



Example — Flipkart: A warehouse can connect its computers to a central switch. This makes faults easier to find and devices easier to add.

Bus: All devices share one main backbone cable.

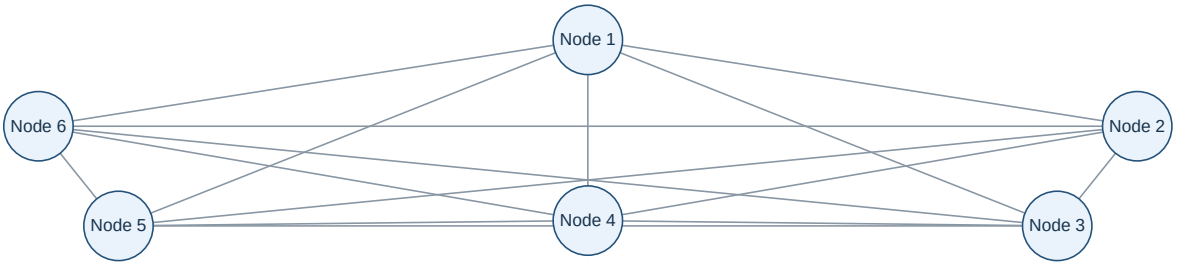
Bus Topology



Example — IPL streaming: Bus is a simple way to demonstrate a shared communication line, although modern streaming services use more complex networks.

Mesh: Devices have multiple connections and more than one possible data path.

Mesh Topology



Example — WhatsApp: Internet traffic can use alternate routes between networks, similar to the reliability idea behind mesh topology.

2. Comparison Table: Star, Bus, Ring, Mesh

Topology	Pros (4)	Cons (4)	Best use case / app example
Star	Easy to install; easy to troubleshoot; one cable failure affects one device; easy to expand.	Switch failure can stop network; more cable; central switch costs money; switch capacity can limit performance.	Small offices, colleges, warehouses. Example: Flipkart-style order office.
Bus	Less cable; low initial cost; simple for small setup; no central switch required.	Backbone failure affects all; hard to troubleshoot; slows with heavy traffic; limited length/devices.	Small or legacy networks/learning labs. Example: shared-line network demonstration.
Ring	Orderly data path; predictable flow; can reduce collisions in some designs; some systems regenerate signals at each device.	A break can disrupt a basic ring; adding devices may interrupt service; faults can be hard to locate; data may pass through several devices.	Some industrial or legacy networks. Example: a hypothetical office network supporting Zomato operations.
Mesh	Multiple paths; highly reliable; can survive some link failures; traffic can reroute.	Expensive; many links and ports needed; complex to manage; full mesh is hard to scale.	Network backbone or wireless mesh. Example: resilient Internet routes supporting Instagram/Paytm services.

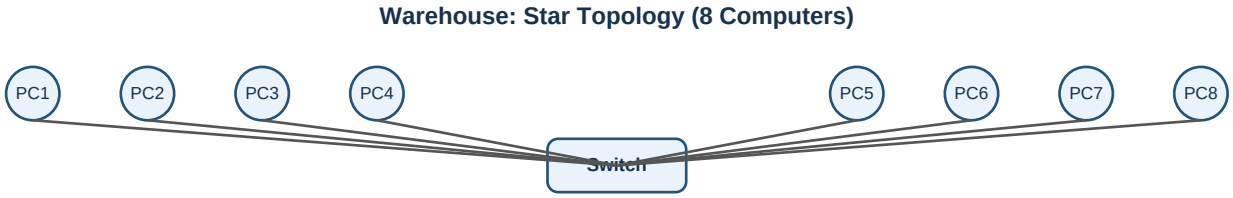
Note: App examples are illustrative; they do not mean that the named app uses only that topology.

3. Best Topology for a Multiplayer Mobile Game

I would choose a **client-server design with reliable, redundant network paths** for a game like PUBG or Free Fire. Players connect to nearby game servers, reducing delay (ping) and improving speed. Backup links and servers improve reliability if a connection fails. This costs more than a simple setup, but fast response and stable play are important for multiplayer games.

4. Warehouse with 8 Computers

Recommended topology: **Star topology**. Connect all eight computers to one central network switch.



Star is easy to install, troubleshoot, and expand. If one computer cable fails, the other computers can keep working. The switch costs extra, so choose a reliable switch.

Conclusion

Every topology has advantages and disadvantages. Star is practical for small offices, while mesh is useful when alternate paths and reliability are important.